
Grace Hopper at a UNIVAC-1 console

house for J Lyons and later for other companies like Ford UK. A later model, the LEO III, completed in 1961 used core memory and employed concurrent programming—running as many as 16 applications simultaneously. The LEO III remained in service until 1981 with British Telecom.

1952
The Compiler

June: A general-purpose, von Neumann computer

The IAS General Purpose Computer was completed by von Neumann and his team at the Institute of Advanced Studies, Princeton. Work on the IAS started around 1946 and the project had a budget of several hundred thousand dollars (in the order of a crore of rupees

Founding Father

John von Neumann, a Hungarian mathematician (who took American citizenship), was one of the greatest mathematicians in history. He made outstanding contributions in numerous areas of physics, mathematics, economics, and computer science. Among other things he is known for the computer architecture which bears his name—a control unit, an arithmetic and logical unit, memory, and I/O. The von Neumann architecture is at the heart of the design of all modern-day computers—from handhelds to supercomputers. An alternative architecture, known as the Harvard architecture, where the instructions and data are separately stored, is used in some components (digital signal processors used in audio or video signal processing). These types of machines can read data and instructions simultaneously, unlike von Neumann machines.